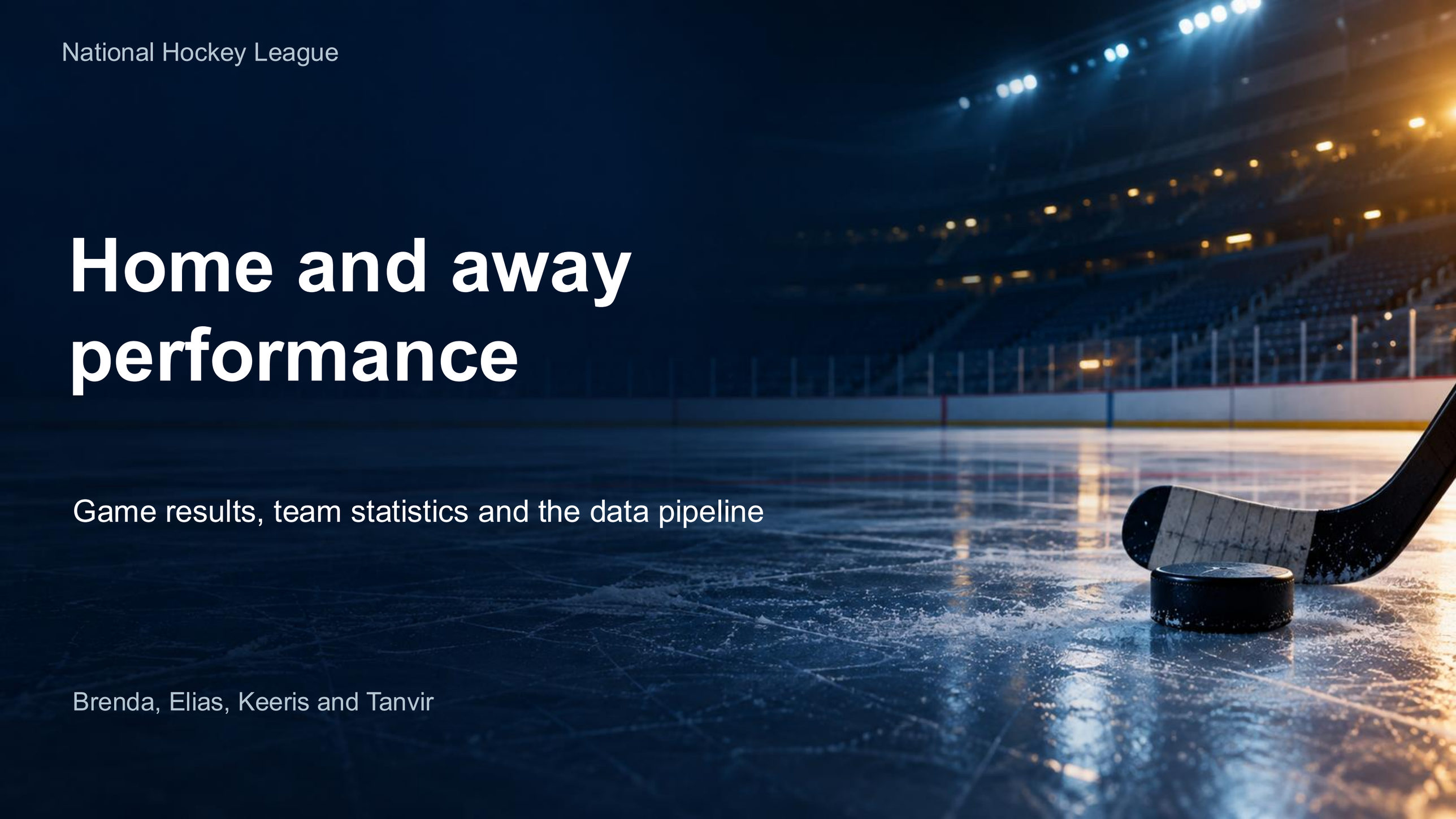
National Hockey League

Home and away performance

Game results, team statistics and the data pipeline

Brenda, Elias, Keeris and Tanvir



Agenda

- Overview
- Data Source
- Tools & Data Architecture
- Challenges
- Data Analysis & Visualizations
- Limitations
- Areas for Improvement

Project questions and approach

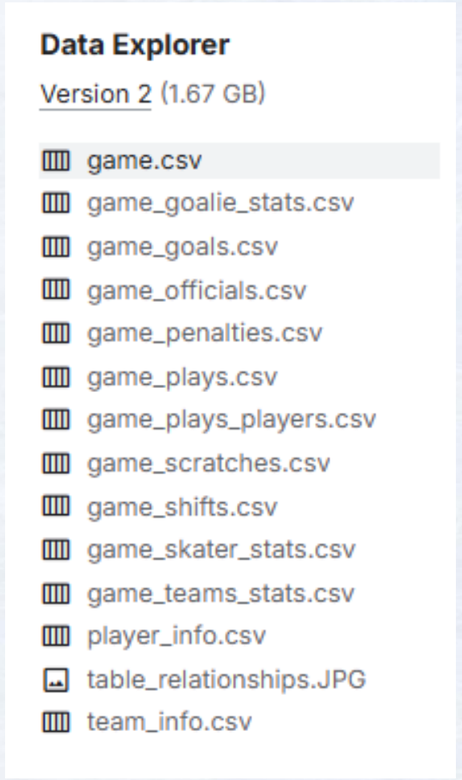
Home advantage	How do home and away teams differ in win rates, scoring, physical play and puck management?
Variation	How does the home advantage gap change across teams, seasons and competition types?
Delivery	Build and validate the NHL data pipeline in Microsoft Fabric, then present the analysis in Power BI.

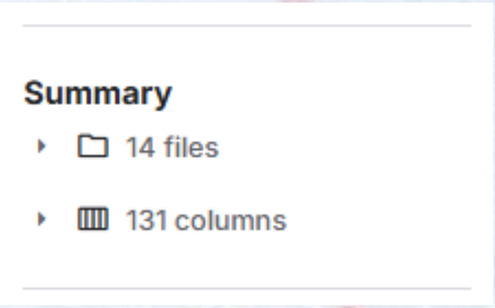
Dataset and analysis scope

Kaggle NHL Game Data

23,716 games in the report
Seasons 2000/01 to 2019/20
628 historical tied games retained

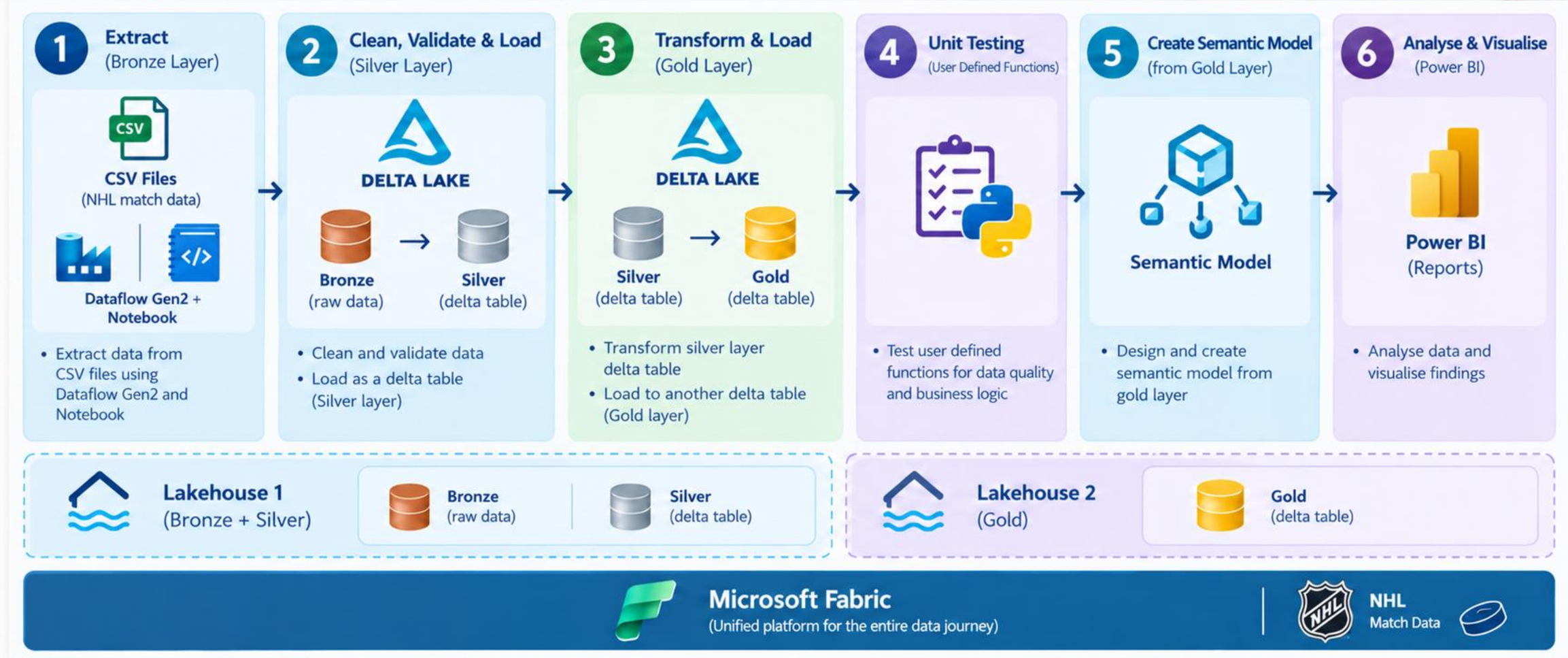
The source includes game results, team statistics, player statistics and play-level records.





Source: kaggle.com/datasets/martinellis/nhl-game-data

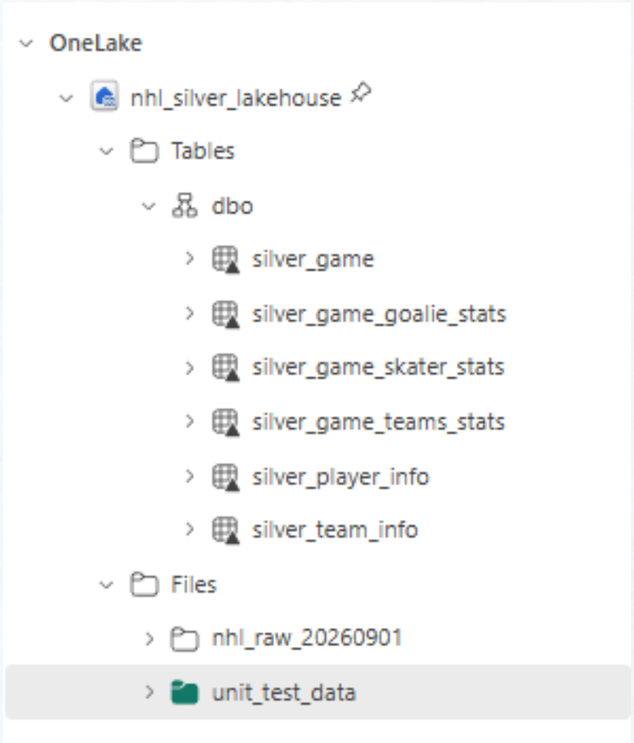
Data Architecture



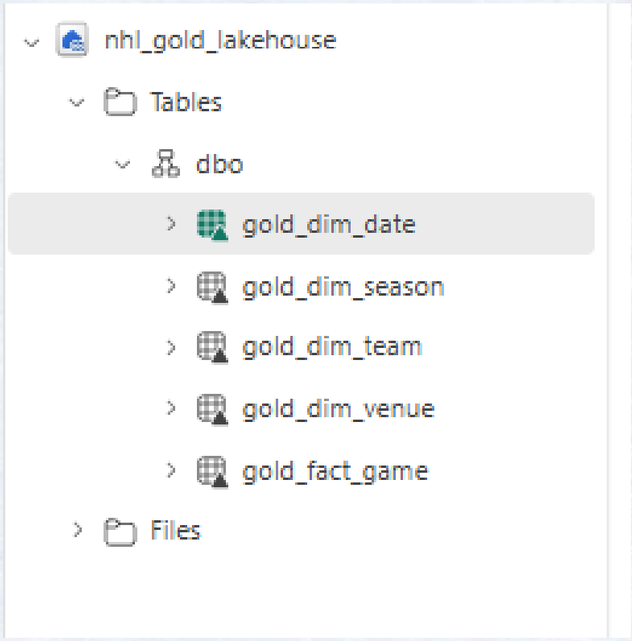
Upload CSV files to Lakehouse. Dataflow Gen2, PySpark, Spark SQL, Python and SQL support ingestion, preparation and validation. Power BI presents the results.

Lakehouse layers

Silver: cleaned tables

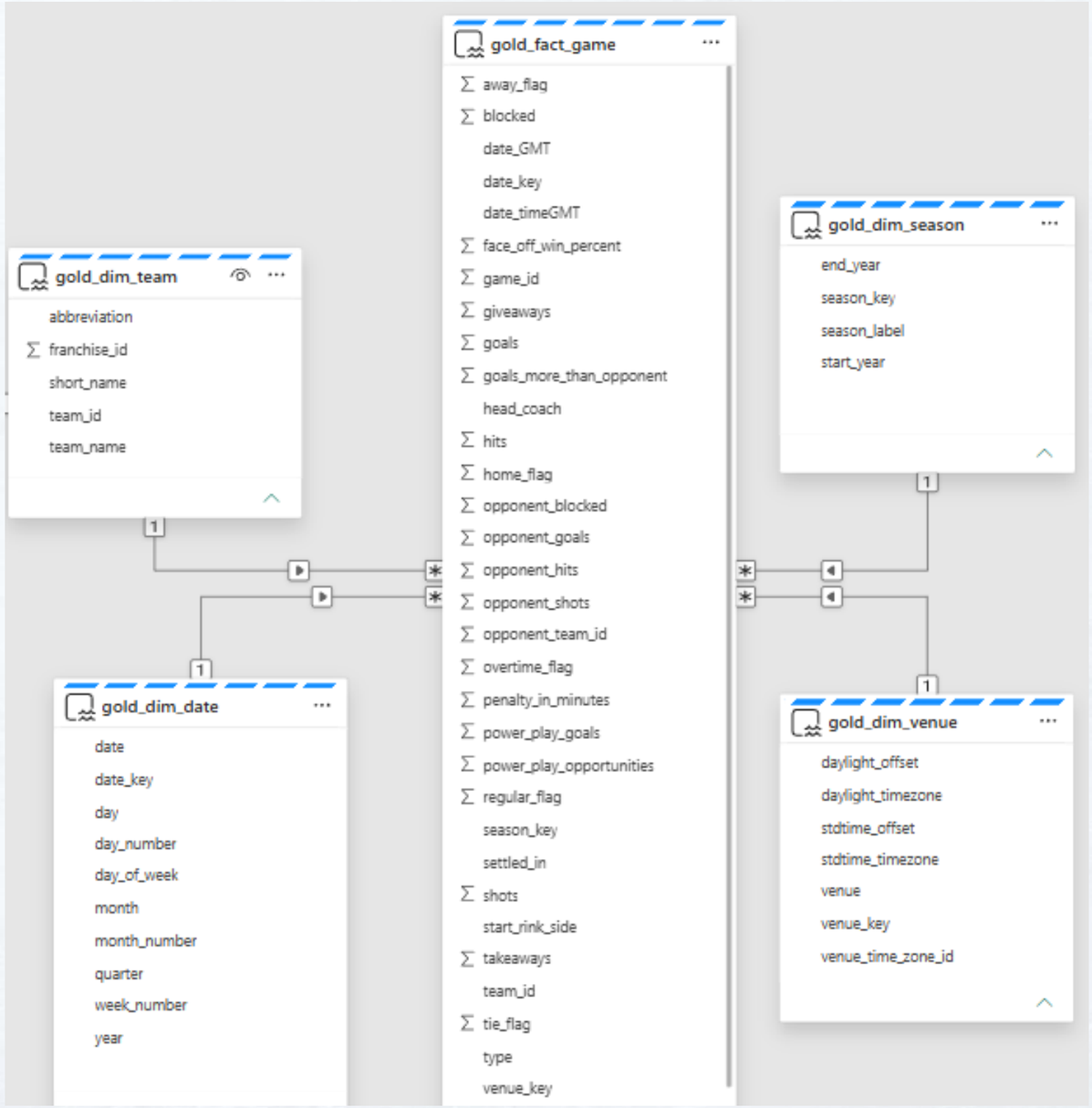


Gold: analysis model



The project separates cleaned source tables from dimensions and the game fact table used for reporting.

Semantic model



The game fact table connects to date, season, team and venue dimensions.

Building Data quality into the Silver Layer

Raw Data to Silver Tables

Raw → Profile → Clean → Validate → Quarantine → Silver

Data quality and Validation Workflow

Profile and standardize	Profile schemas, duplicates and missing-value patterns. Standardize "NA" to spark "NULL" where appropriate and validate data types.
Validate record quality	Validate ranges, categorical values and game-level logic. Cross-check home/away records, goals.
Validate relational integrity	Check PK uniqueness and FK relationships. Investigate unmapped IDs and quarantine genuine orphan records.
Investigate anomalies	Analyse patterns by season, game type, venue and context. Separate legitimate historical gaps from genuine errors.
Publish silver tables	Write cleaned Delta tables to Silver and rerun schema, row-count, uniqueness and PK-FK checks

Challenge: "NA" was not a true NULL

- Some missing values were stored as the literal string "NA".
- Spark treated "NA" as a real value, not a missing value.
- Affected data quality checks and could produce misleading validation results.
- Standardized appropriate "NA" values to Spark NULL.
- Rechecked schemas and missing-value counts after transformation.

Key lesson: Missing data can have several representations.

Challenge: Unrecognized Team IDs & Mapping them

- PK–FK validation initially identified team IDs **87–90** as missing from team_info.
- Investigation showed they represented **NHL All-Star teams**, not erroneous records.
- Added the four team IDs to team_info rather than deleting the associated games.
- This preserved relationships across other related tables.
  For example:game, game_teams_stats, game_skater_stats, game_goalie_stats and other related tables.

Result: Referential integrity restored with **no unnecessary data loss**.

```
Original team_info rows: 33
Cleaned team_info rows: 37
```

Challenge: Orphan Records

- PK-FK validation found **11 distinct play_id values** in game_plays_players that did not exist in game_plays.
- These 11 orphan IDs affected **19 child records**.
- Rather than silently deleting them, the affected records were moved to **quarantine**.
- Revalidated the cleaned table after quarantine.

Result:

```
Invalid play_id after quarantine: 0
```

```
Quarantined rows: 19
```

Challenge: Understanding Historical Missing Data

- Identified systemic historical patterns.
- Distinguished structural missingness from corrupted data.
- Preserved legitimate NULLS where appropriate

Key lesson: Missing values are **not always** errors.
NULL ≠ bad data automatically.

Field	Missing values observed
goals	8
shots	8
pim	8
powerPlayOpportunities	8
powerPlayGoals	8
hits	4,928
giveaways	4,928
takeaways	4,928
blocked	4,928
faceOffWinPercentage	22,148
head_coach	28
startRinkSide	2,198

Challenge: Semantic Quality

- Investigated more subtle data-quality issue.
- Found inconsistent metadata in games table – Venues with different timezones
- Make sense of the data beyond technical validation.

	venue	timezone_count
1	Rogers Place	8
2	Scotiabank...	3
3	Ericsson Gl...	3
4	Enterprise ...	2
5	Hartwall Ar...	2
6	Globe Arena	2
7	O2 Czech R...	2
8	SAP Center...	2
9	Hartwall Ar...	2

Key lesson: Data can be technically valid but still be semantically inconsistent.

game_id	date_time_GMT	away_team_id	home_team_id	venue	venue_time_zone_id	venue_time_zone_offset	venue_time_zone_tz
2019030321	2020-09-07 00:00:00	25	54	Rogers Place	America/Los_Angeles	-8	PST
2019030311	2020-09-08 00:00:00	2	14	Rogers Place	America/New_York	-5	EST
2019030322	2020-09-09 00:00:00	25	54	Rogers Place	America/Los_Angeles	-8	PST
2019030312	2020-09-10 00:00:00	2	14	Rogers Place	America/New_York	-5	EST
2019030323	2020-09-11 00:00:00	54	25	Rogers Place	America/Chicago	-6	CST

Challenge: Profiling and explicit missing-value checks

Overview

Overview | Alerts 10 | Reproduction

Dataset statistics

Number of variables	17
Number of observations	52610
Missing cells	22148
Missing cells (%)	2.5%
Duplicate rows	5140
Duplicate rows (%)	9.8%

Variable types

Numeric	12
Categorical	4
Boolean	1

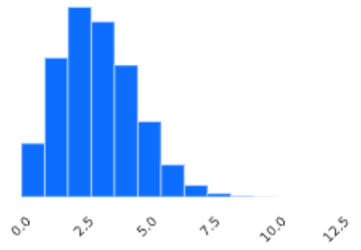
- Struggled to evaluate the data from a bird's eye view in order to determine what needs to be worked on.
- Ydata-profiling summarized distributions and quality alerts. The team supplemented it with column-by-column null checks.

goals

Real number (ℝ)

Zeros

Distinct	13	Minimum	0
Distinct (%)	< 0.1%	Maximum	12
Missing	0	Zeros	3469
Missing (%)	0.0%	Zeros (%)	6.6%
Infinite	0	Negative	0
Infinite (%)	0.0%	Negative (%)	0.0%
Mean	2.7808592	Memory size	0.0 B



More details

Challenge: Data validation on "won" column

- Struggled to validate the data of a column because each game is represented by separate home and away records.

- Solution: Combine and compare both records using game_id and goals column to determine the actual game result and validate whether won column was correctly recorded.

game_id	team_id	home_or_away	won	settled_in	head_coach	goals	game_result
2000020001	21	away	false	OT	Bob Hartley	2	tie
2000020001	25	home	false	OT	Ken Hitchcock	2	tie
2000020002	9	away	false	OT	Jacques Martin	4	tie
2000020002	6	home	false	OT	Pat Burns	4	tie
2000020003	16	away	false	REG	Alpo Suhonen	2	lose
2000020003	7	home	true	REG	Lindy Ruff	4	win

Silver to Gold Table Transformation

SparkSQL --> Physical Table --> Validate & Log with PySpark

Challenge: Venue and Seasonal Time Changes

Standard and daylight offsets produced multiple rows for a venue.

- Solution: Pivoted into separate columns – (1) standard time (2) daylight time

	ABC venue	ABC venue_time_zone_id	ABC venue_time_zone_tz	123 venue_time_zone_offset
1	Air Canada Centre	America/Toronto	EDT	-4
2	Air Canada Centre	America/Toronto	EST	-5
3	Amalie Arena	America/New_York	EST	-5
4	Amalie Arena	America/New_York	EDT	-4

Pivoting resulted in nulls in time zone columns for venue that were not used during the period.

- Solution: Filling nulls with derived offsets from either columns. Phoenix separately hard coded.

123 venue_key	ABC venue	ABC venue_time_zone_id	ABC st_tz	123 st_tz_offset_redef	ABC dt_tz	123 dt_tz_offset_redef
1	Air Canada Centre	America/Toronto	EST	-5	EDT	-4
2	Amalie Arena	America/New_York	EST	-5	EDT	-4
3	America West Arena	America/Phoenix	MST	-7	*NULL*	*NULL*
4	American Airlines Center	America/Chicago	CST	-6	CDT	-5
5	Arrowhead Pond	America/Los_Angeles	*NULL*	*NULL*	PDT	-7
6	BB&T Center	America/New_York	EST	-5	EDT	-4

Challenge: Data Quality Issue with Venue & Timezone ID

Some venues are associated with several time-zone IDs

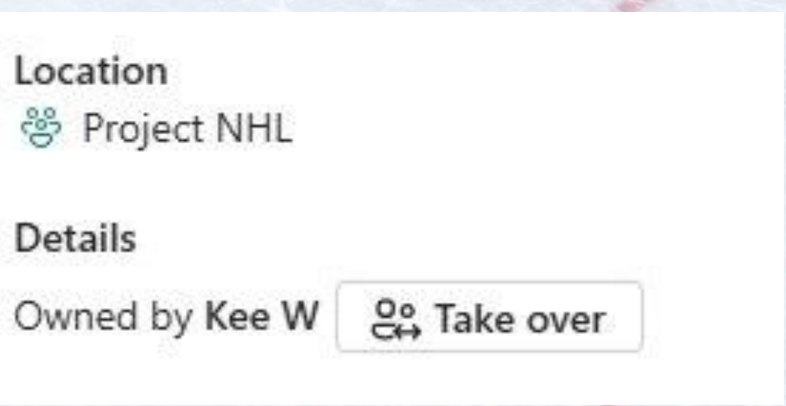
- Joining on dim_venue alone duplicated rows in the game fact table (not unique)
- Solutions:
  - (1) Select 1 time-zone ID for each venue, using frequency-based approach.
  - **(2) Use of composite keys – venue / time-zone id**
- Added surrogate key - venue_key for future use
- Relied on row count validation before and after transformation

	123 ven...	ABC venue	ABC venue_time_zone_id
26	26	Dodger Stadium	America/Los_Angeles
27	27	Enterprise Center	America/New_York
28	28	Enterprise Center	America/Chicago
29	29	Entertainment and Sports Arena	America/New_York
30	30	Ericsson Globe	America/Los_Angeles
31	31	Ericsson Globe	America/Denver
32	32	Ericsson Globe	America/New_York
33	33	Falcon Stadium	America/Denver

Challenge: Semantic Model Access

Data model does not automatically refresh

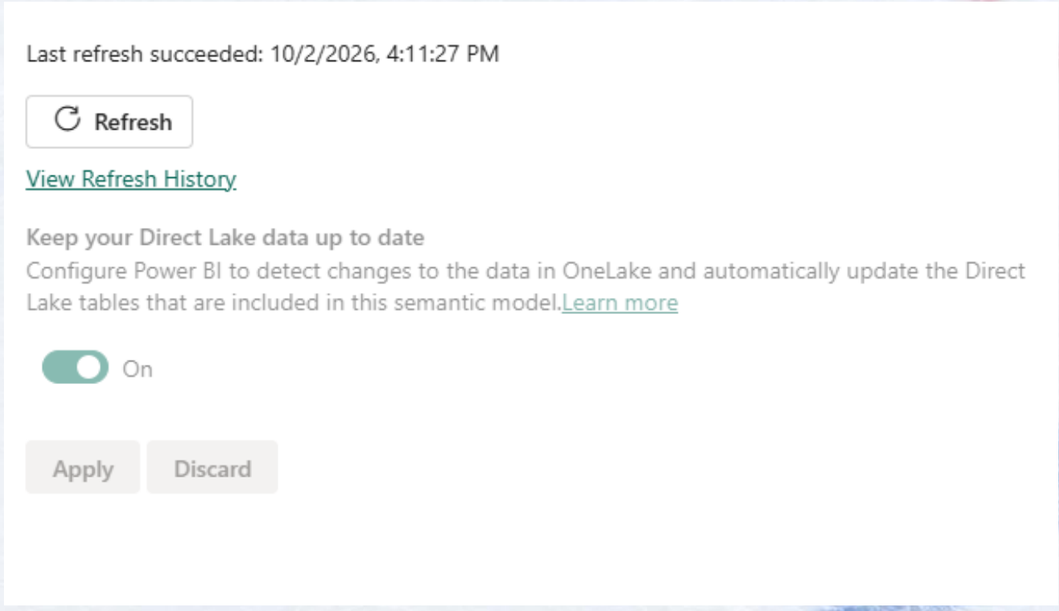
- Changes made to gold tables cannot be reflected until data model owner refreshes
- Solution: Schedule refresh by Data Model Owner



Other Challenges

Hockey terminology and analytics

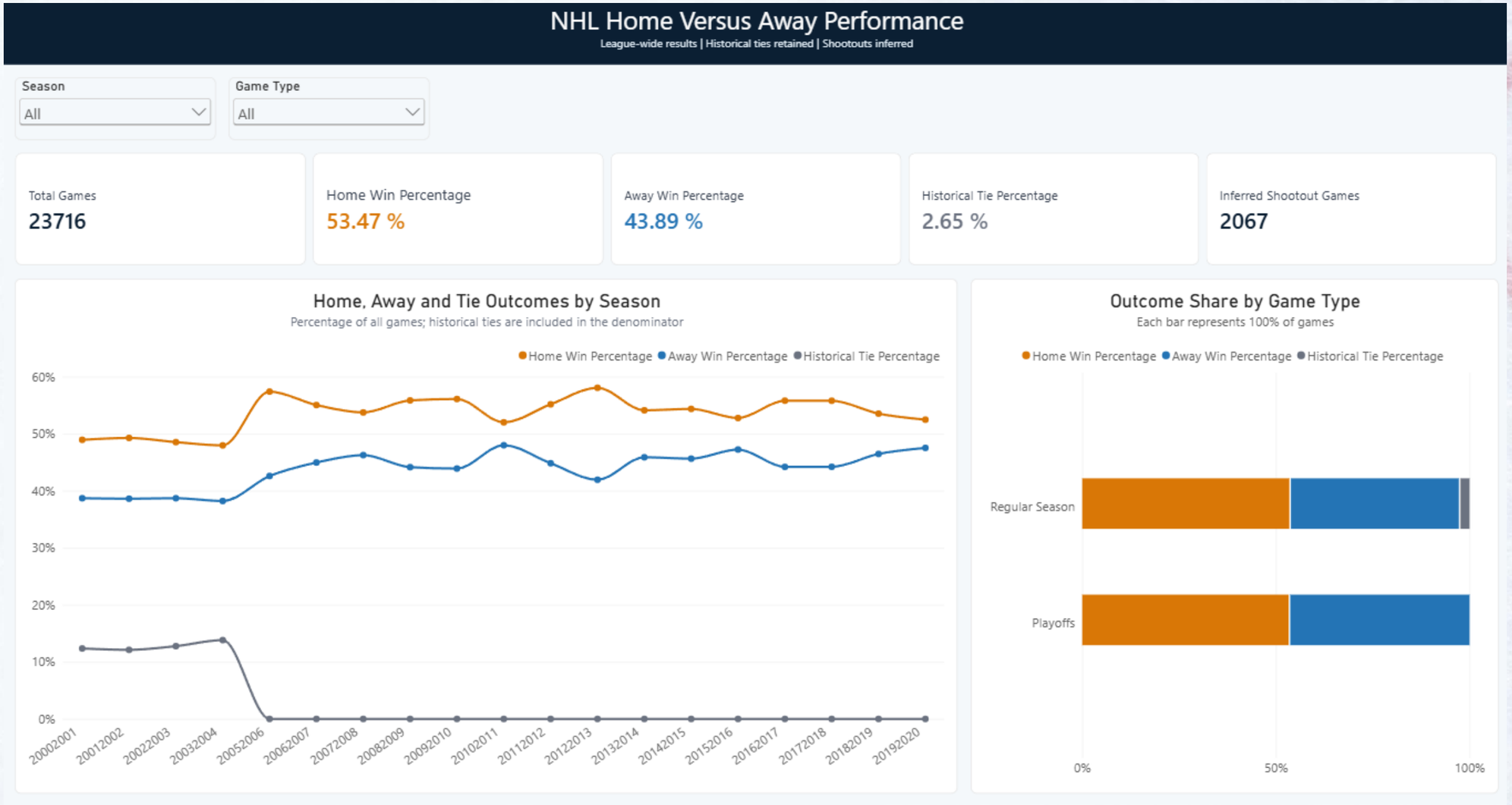
- Understanding the purpose of the different metrics for a meaningful analytics



Metric definitions and comparison rules

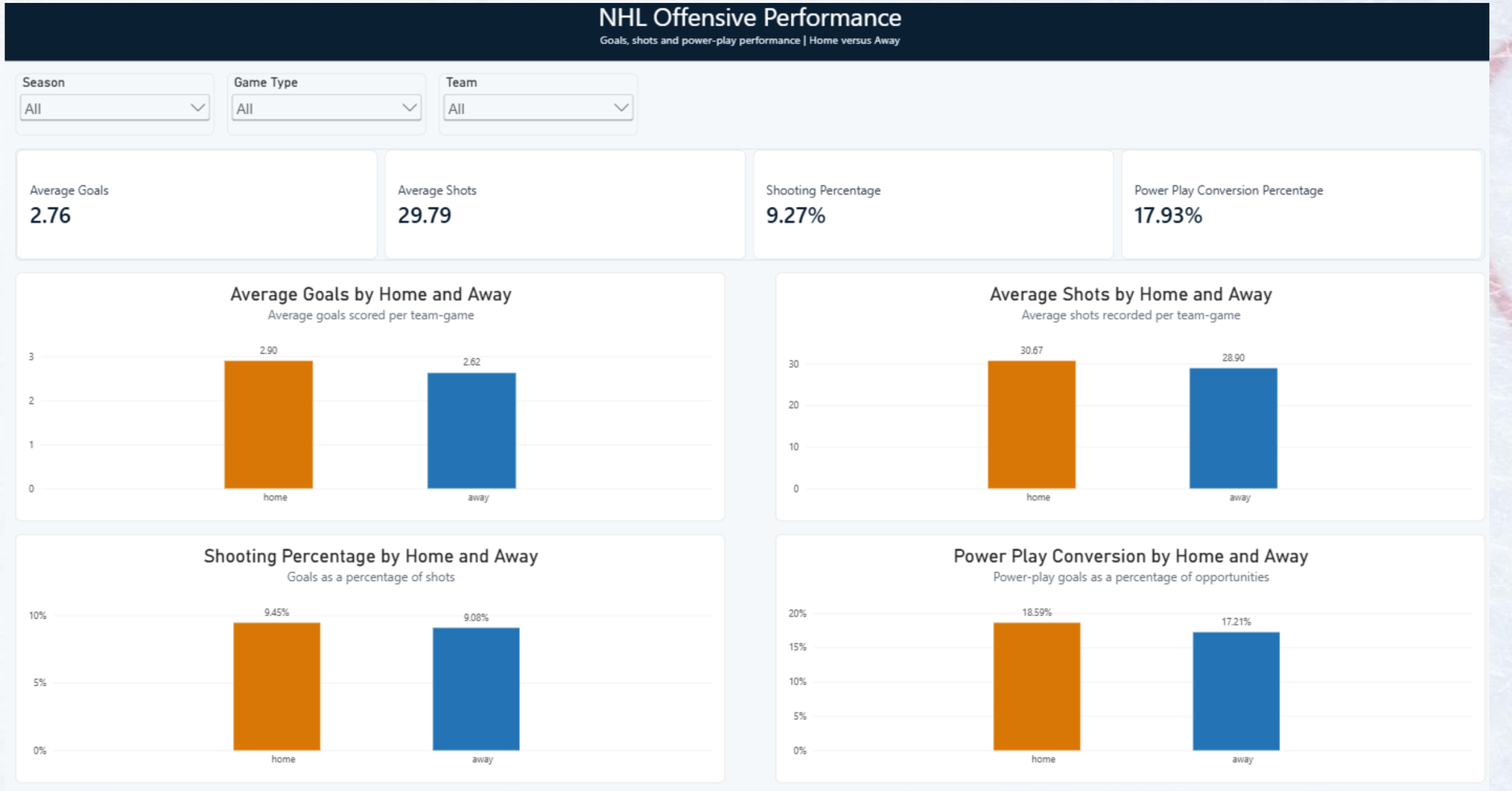
Match	• Game Type: Regular, Playoff, All-Star • Settled In: Regular, Overtime • Team: Home, Away
Scoring	• Goals • Shots • PowerPlay opportunities, PowerPlay goal
Physical play	• Penalty minutes (PIM) : Total time spent in penalty box • Hits: Physical contact separating opponent from puck • Blocked : Stopping opponent's shot with body/stick • Giveaways : careless loss that gifts free score to opposing team • Takeaways : Stealing puck possession from opponent • Faceoff win percentage: Ratio of won puck drop during faceoffs

Home teams win more often at league level



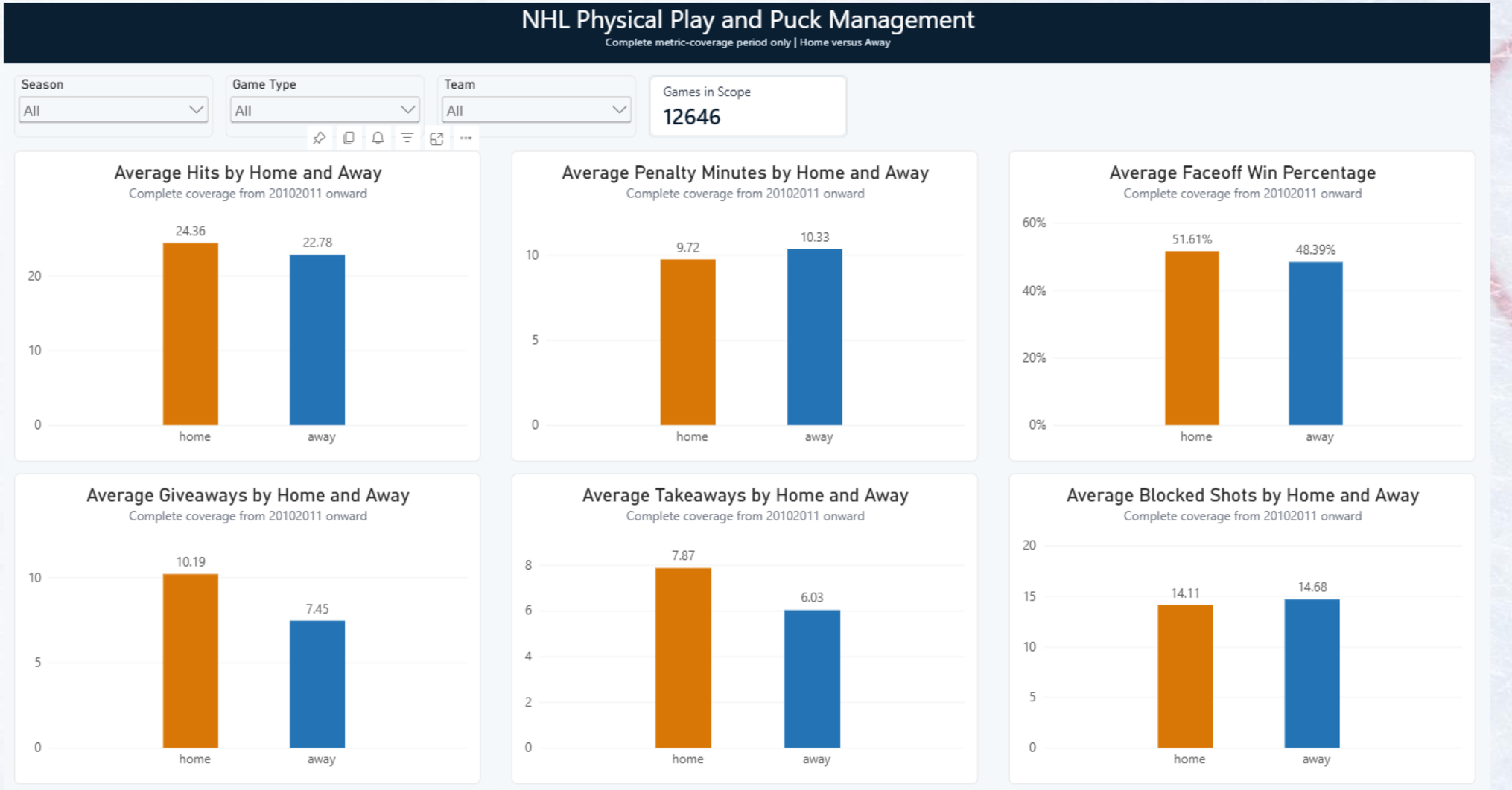
Home wins: 53.47%. Away wins: 43.89%. Gap: 9.58 percentage points across 23,716 games.

Home teams score more and convert more chances



Home versus away: 2.90 vs 2.62 goals per team-game and 18.59% vs 17.21% power-play conversion.

Physical-play differences are mixed



Home teams record more hits and takeaways, fewer penalty minutes, and more giveaways. Scope: 12,646 games.

Home advantage varies across teams and seasons

NHL Team Home Advantage

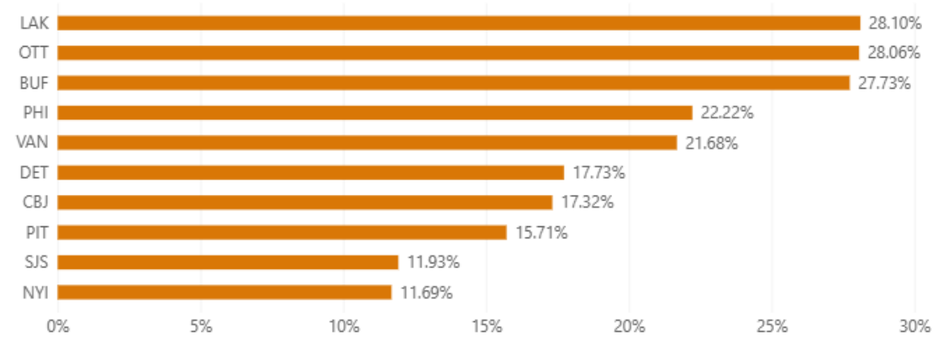
Team-season comparison | Minimum 20 home and 20 away games

Season: 20192020

Regular Season only | Teams require at least 20 home and 20 away games

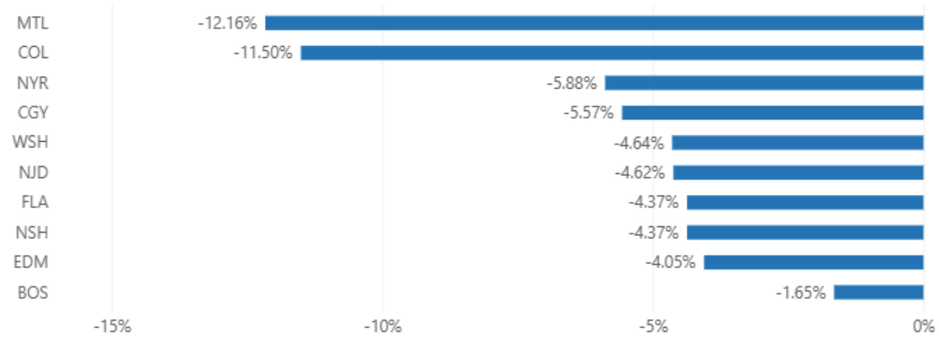
Largest Home Advantage Gaps

Home win % minus away win %



Smallest Home Advantage Gaps

Negative values indicate stronger away performance



Eligible Team Detail

team_abbreviation	Team Home Games	Team Away Games	Team Home Win Percentage	Team Away Win Percentage	Team Home Advantage Gap
LAK	34	36	55.88%	27.78%	28.10%
OTT	37	34	48.65%	20.59%	28.06%
BUF	35	34	57.14%	29.41%	27.73%
PHI	36	36	72.22%	50.00%	22.22%
VAN	35	34	62.86%	41.18%	21.68%
DET	37	34	32.43%	14.71%	17.73%
CBJ	36	34	55.56%	38.24%	17.32%
PIT	35	34	65.71%	50.00%	15.71%
Total	**1094**	**1094**	**53.20%**	**46.80%**	**6.40%**

2019/20 regular season: LAK +28.10 points, MTL −12.16 points. Each eligible team has at least 20 home and 20 away games.

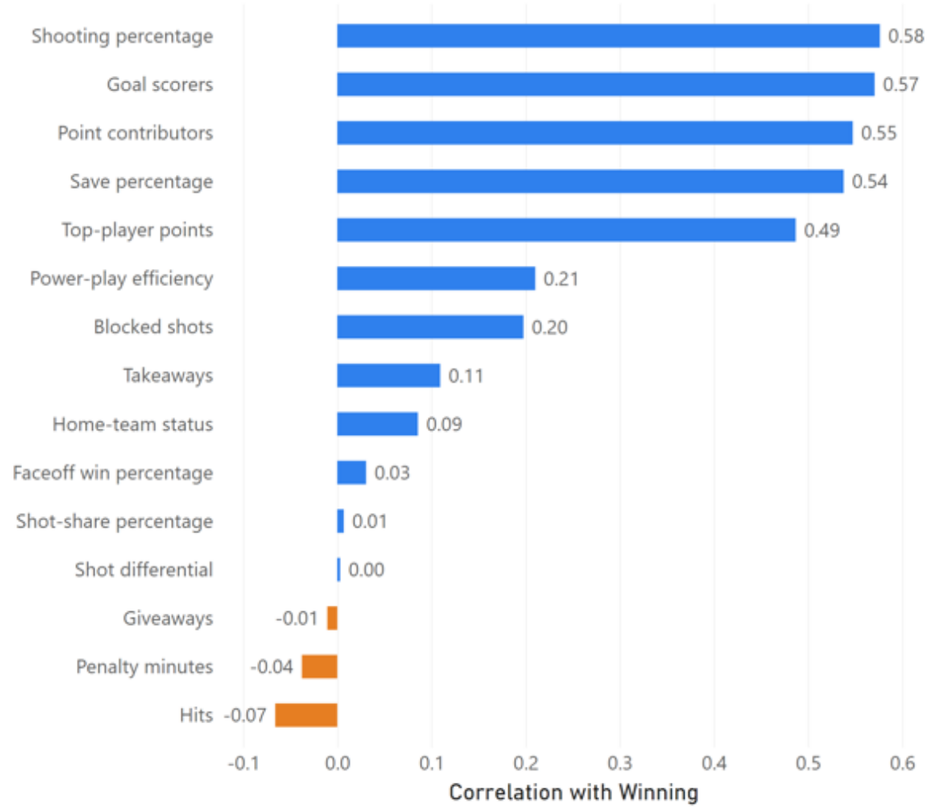
Shooting percentage has the strongest reported correlation

Win-Factor Overview

Performance factors associated with winning NHL games

Analysis window: 2010/11–2019/20

Win Factors Ranked by Correlation



How Winners Differ from Losers

Metric	Winner Avg	Loser Avg	Winner–Loser Gap
Blocked shots	15.41	13.39	2.01
Faceoff win percentage	50.22	49.78	0.44
Giveaways	8.77	8.87	-0.10
Goal scorers	3.29	1.67	1.61
Hits	23.00	24.14	-1.14
Home-team share	0.54	0.46	0.09
Penalty minutes	9.73	10.32	-0.60
Point contributors	7.42	4.18	3.24
Power-play efficiency	23.94	13.47	10.46
Save percentage	94.00	88.03	5.97
Shooting percentage	12.72	6.01	6.71
Shot difference	0.03	-0.03	0.06
Shot share percentage	50.06	49.94	0.12
Takeaways	7.36	6.53	0.83
Top player points	2.17	1.27	0.90

Special-teams note:
Power-play and penalty-kill percentages mirror each other, so they are treated as one special-teams factor.

Key takeaway:
Shooting percentage has the strongest link to winning (r = 0.58), followed by goal scorers, point contributors, and save percentage. Other metrics show weak correlations. These are associations, not proof of causation.

2010/11–2019/20: shooting percentage r = 0.58 and home-team status r = 0.09.

Interpretation and limitations

Association and causation

Venue comparisons do not isolate venue effects. Opponent strength, rosters, scheduling and historical changes may influence results.

Data coverage

Older seasons lack several metrics. The supplied files exclude injuries, travel distance, rest days and attendance.

Win-factor method

Same-game scoring measures relate directly to the result. Correlation formulas and uncertainty estimates need verification.

Next steps for analysis and delivery

Stronger comparisons

Use scatter plots and confirm metric formulas. Assess opponent strength, rest and travel with additional data.

Automated quality checks

Add repeatable checks for duplicate game–team rows, unmatched IDs, missing statistics, inconsistent results and duplicate joins.

Reliable operation

Schedule notebooks with failure alerts and version shared work. Agree access rights and model ownership. Coordinate gold-layer changes with semantic-model updates.

Home advantage varies

9.58

percentage-point gap in league win rates

The gap changes across teams and seasons and can reverse for individual teams.

The project supports a descriptive home advantage. Explaining its causes requires further analysis.



Thank You